



Cloud Computing

BSE-VB

Submitted By

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Submitted to

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Open-ended Lab

Phase 1 – Terraform Infrastructure Setup

- Goal: Provision VPC, subnet, security groups, and EC2 instances.
- Steps:
 - Wrote main.tf, variables.tf, outputs.tf.
 - Defined 3 backend EC2s + 1 frontend EC2.
 - Applied terraform init → terraform apply.
- Screenshots:

environment_versions.png:

```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform --version
_
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ aws --version
aws-cli/2.33.1 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/site-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
```

terraform_init.png:

```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/http from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/http v3.5.0
- Using previously-installed hashicorp/aws v6.28.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

terraform_apply.png:

```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform apply -auto-approve
data.http.my_ip: Reading...
data.http.my_ip: Read complete after 1s [id=https://icanhazip.com]
aws_vpc.this: Refreshing state... [id=vpc-01027b7af0b5dc770]
aws_internet_gateway.this: Refreshing state... [id=igw-03edf29190b7dcf33]
aws_subnet.public: Refreshing state... [id=subnet-004de94a8d383e876]
aws_security_group.web_sg: Refreshing state... [id=sg-0445f9aa98f2fa0b7]
aws_route_table.public: Refreshing state... [id=rtb-0eb06675158d6650a]
aws_route_table_association.public: Refreshing state... [id=rtbassoc-0a03834579cd2c737]
aws_instance.backend[2]: Refreshing state... [id=i-0da535d85c7cade1a]
aws_instance.backend[0]: Refreshing state... [id=i-00452d4ff96debf0]
aws_instance.frontend: Refreshing state... [id=i-077cf4da56fcdceef]
aws_instance.backend[1]: Refreshing state... [id=i-013be12a7ea7cee5c]

Terraform used the selected providers to generate the following execution plan. Resource actions are
indicated with the following symbols:
  ~ update in-place
```

```
# aws_instance.backend[2] will be created
+ resource "aws_instance" "backend" {
  + ami                  = "ami-057e2bc910cc38f26"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone      = (known after apply)
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized          = (known after apply)
  + enable_primary_ipv6    = (known after apply)
  + force_destroy          = false
  + get_password_data       = false
  + host_id                = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                    = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count       = (known after apply)
```

```
# aws_instance.frontend will be created
+ resource "aws_instance" "frontend" {
  + ami                                = "ami-057e2bc910cc38f26"
  + arn                                = (known after apply)
  + associate_public_ip_address       = (known after apply)
  + availability_zone                 = (known after apply)
  + disable_api_stop                   = (known after apply)
  + disable_api_termination           = (known after apply)
  + ebs_optimized                     = (known after apply)
  + enable_primary_ipv6               = (known after apply)
  + force_destroy                     = false
  + get_password_data                 = false
  + host_id                           = (known after apply)
  + host_resource_group_arn           = (known after apply)
  + iam_instance_profile              = (known after apply)
  + id                                = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle                = (known after apply)
  + instance_state                    = (known after apply)
  + instance_type                     = "t3.micro"
  + ipv6_address_count                = (known after apply)
  + ipv6_addresses                    = (known after apply)
  + key_name                          = "Lab10user"
```

Apply complete! Resources: 4 added, 1 changed, 4 destroyed.

Outputs:

```
backend_private_ips = [
  "10.0.1.56",
  "10.0.1.130",
  "10.0.1.48",
]
backend_public_ips = [
  "51.112.48.222",
  "3.29.139.222",
  "40.172.177.131",
]
frontend_public_ip = "51.112.52.209"
```

Ec2_instances.png:

☐	Name 	Instance ID	Instance state 	Instance type 	Status check	Alarm
☐	labproj-backe...	i-05e344c44cdc0d445	 Running  	t3.micro	 3/3 checks passec	View
☐	labproj-frontend	i-0eec4eb156d91915f	 Running  	t3.micro	 3/3 checks passec	View
☐	labproj-backe...	i-0025d0bf882c26b0d	 Running  	t3.micro	 3/3 checks passec	View
☐	labproj-backe...	i-0ed96fc3efde1adc4	 Running  	t3.micro	 3/3 checks passec	View

Phase 2 – Ansible Roles Setup

- Created Ansible roles:
 - backend for HTTPD servers
 - frontend for Nginx load balancer
- Configured ansible.cfg for SSH key and Python interpreter.
- Inventory initially static using Terraform output IPs.

Phase 3 – Backend HTTPD Role

- Backend servers configured to:
 - Install Apache HTTPD
 - Enable & start the service
 - Deploy a unique index page per server showing hostname and private IP
- Verified that each backend shows **distinct content**.
- **Screenshots:**

Backend_server1.png



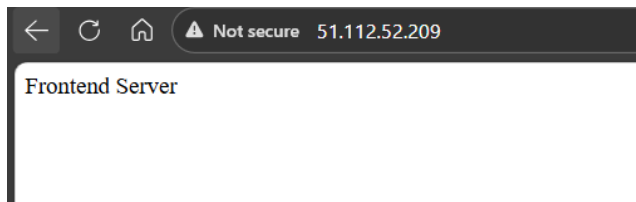
Backend_server2.png



Backend_server3.png



frontend_server1.png



Phase 4 – Frontend Nginx Role

- Frontend server configured to:
 - Install Nginx
 - Deploy Nginx config as **load balancer**
 - Upstream configured with **2 primary backends + 1 backup**
- Initially used public IPs for testing, later switched to private IPs.
- Verified load balancing between primary backends.
- Screenshots:

```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend/ansible (main) $ ansible-inventory -i
inventory/aws_ec2.yaml --graph
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This feature will be removed from ansible
-core version 2.23.
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |   |--labproj-backend-1
  |   |--labproj-backend-2
  |   |--labproj-backend-3
  |   |--labproj-frontend
  |--@backend:
  |   |--labproj-backend-1
  |   |--labproj-backend-2
  |   |--labproj-backend-3
  |--@frontend:
  |   |--labproj-frontend
```

```
backend_public_ips = [
```

```
    "51.112.48.222",
```

```
    "3.29.139.222",
```

```
    "40.172.177.131",
```

```
]
```

```
frontend_public_ip = "51.112.52.209"
```

backend_servers.png:





Frontend_server.png

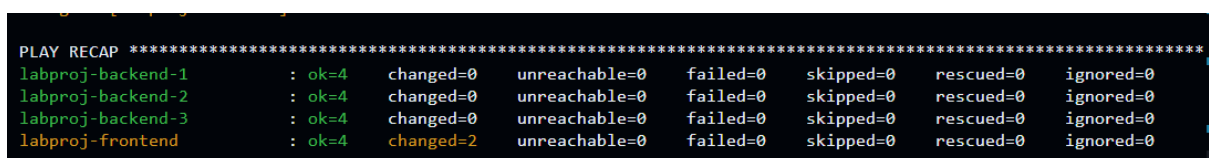
I ran frontend Ip and refresh it to see the backend Ips :



Phase 5 – Main Playbook

- Combined roles in site.yaml:
 - Backend role applied to all backend servers
 - Frontend role applied to frontend server with backend IPs passed as variables
- Verified manually:
 - Backend pages show correct hostname & private IP
 - Frontend alternates traffic between primary backends
 - Backup backend serves if primaries stopped
- **No changes were needed in roles or templates, only the playbook variables.**
- **Screenshots:**

Playbook_output.png:



Test backup behavior

Stop Apache on backend1 and backend2:

bash

ssh ec2-user@<backend1-public-ip> "sudo systemctl stop httpd"

ssh ec2-user@<backend2-public-ip> "sudo systemctl stop httpd"

backup_server.png:



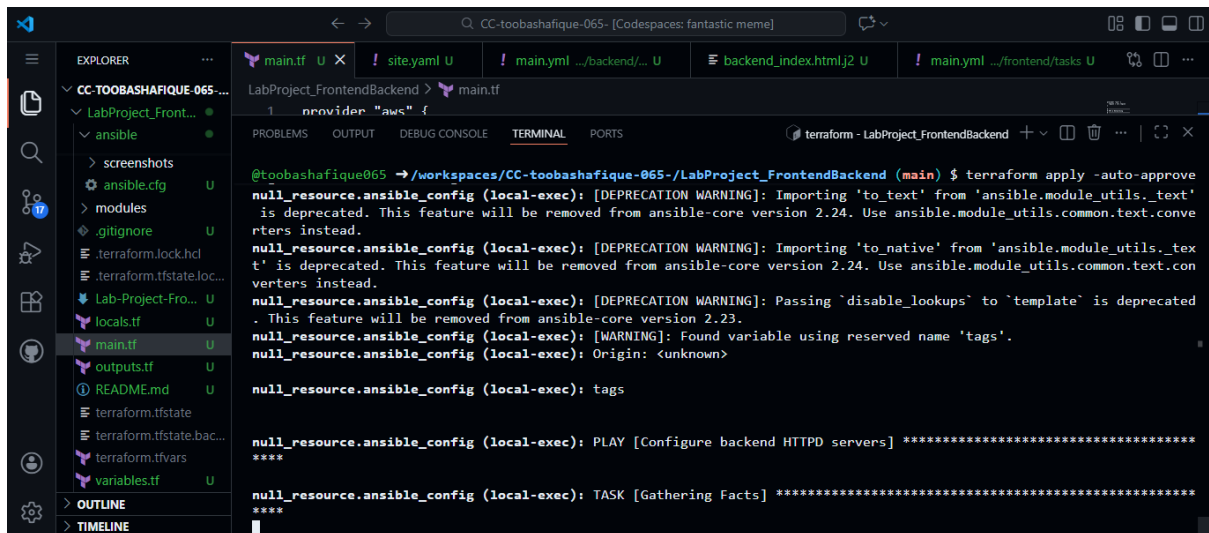
Phase 6 – Terraform + Ansible Automation

- Added Terraform `null_resource` to automatically run Ansible after terraform apply.
- Verified automation:
 - terraform destroy -auto-approve then terraform apply -auto-approve recreates everything
 - No manual ansible-playbook run required
 - Infrastructure and configuration applied correctly and idempotently.
- Screenshots:

Terraform_destroy.png:

```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform destroy -auto-approve
aws_instance.frontend: Still destroying... [id=i-0eec4eb156d91915f, 00m40s elapsed]
aws_instance.backend[0]: Destruction complete after 41s
aws_instance.backend[1]: Destruction complete after 41s
aws_internet_gateway.this: Still destroying... [id=igw-03edf29190b7dcf33, 00m40s elapsed]
aws_instance.backend[2]: Still destroying... [id=i-0ed96fc3efde1adc4, 00m50s elapsed]
aws_instance.frontend: Still destroying... [id=i-0eec4eb156d91915f, 00m50s elapsed]
aws_internet_gateway.this: Destruction complete after 49s
aws_instance.frontend: Destruction complete after 51s
aws_instance.backend[2]: Destruction complete after 52s
aws_subnet.public: Destroying... [id=subnet-004de94a8d383e876]
aws_security_group.web_sg: Destroying... [id=sg-0445f9aa98f2fa0b7]
aws_subnet.public: Destruction complete after 0s
aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.this: Destroying... [id=vpc-01027b7af0b5dc770]
aws_vpc.this: Destruction complete after 0s
Destroy complete! Resources: 10 destroyed.
```

Terraform_apply.png:



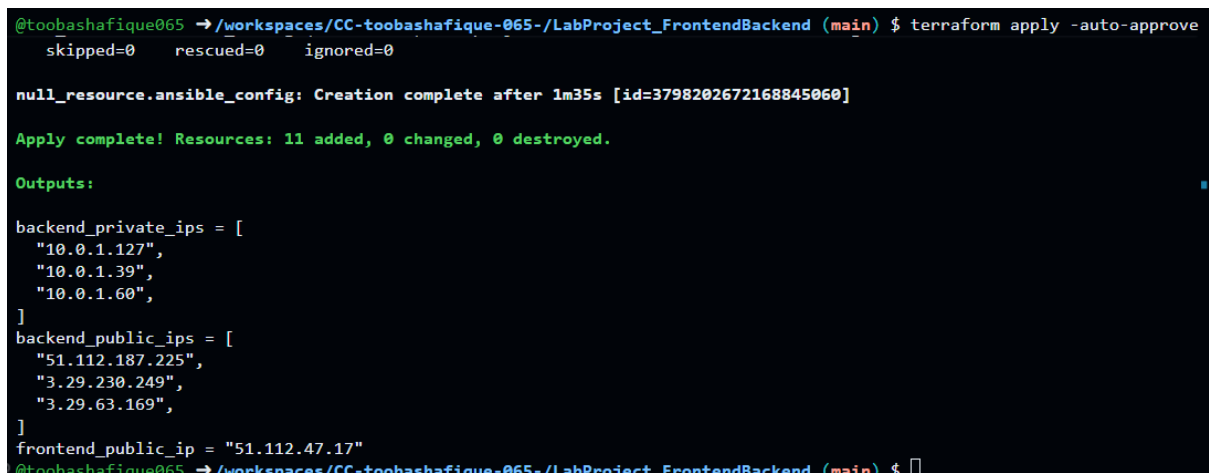
```
LabProject_FrontendBackend > main.tf
1 provider "aws" {

@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform apply -auto-approve
null_resource.ansible_config (local-exec): [DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils.text'
is deprecated. This feature will be removed from ansible-core version 2.24. Use ansible.module_utils.common.text.conve
rters instead.
null_resource.ansible_config (local-exec): [DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._tex
t' is deprecated. This feature will be removed from ansible-core version 2.24. Use ansible.module_utils.common.text.con
verters instead.
null_resource.ansible_config (local-exec): [DEPRECATION WARNING]: Passing 'disable_lookups' to 'template' is deprecated
. This feature will be removed from ansible-core version 2.23.
null_resource.ansible_config (local-exec): [WARNING]: Found variable using reserved name 'tags'.
null_resource.ansible_config (local-exec): Origin: <unknown>

null_resource.ansible_config (local-exec): tags

null_resource.ansible_config (local-exec): PLAY [Configure backend HTTPD servers] *****
****

null_resource.ansible_config (local-exec): TASK [Gathering Facts] *****
****
```



```
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $ terraform apply -auto-approve
skipped=0    rescued=0    ignored=0

null_resource.ansible_config: Creation complete after 1m35s [id=3798202672168845060]

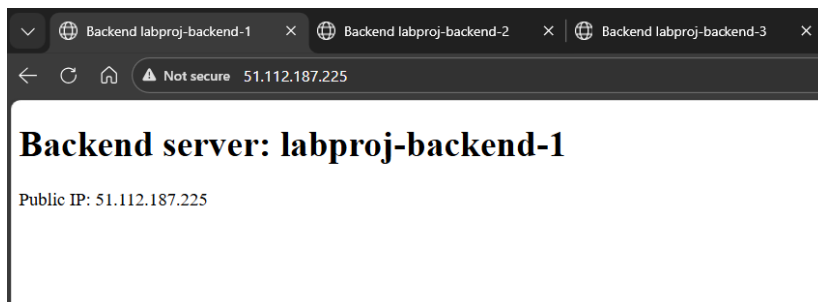
Apply complete! Resources: 11 added, 0 changed, 0 destroyed.

Outputs:

backend_private_ips = [
  "10.0.1.127",
  "10.0.1.39",
  "10.0.1.60",
]
backend_public_ips = [
  "51.112.187.225",
  "3.29.230.249",
  "3.29.63.169",
]
frontend_public_ip = "51.112.47.17"
@toobashafique065 → /workspaces/CC-toobashafique-065-/LabProject_FrontendBackend (main) $
```

Backend_servers.png:

All 3 backend servers and 1 frontend server ran:



Frontend_server.png:

When I refresh it again and again it shows backend servers one by one.

